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PROJECT-2 REPORT

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Project Topic

Machine Learning and Genetic Algorithm Based Personalized Nutrition Plan Generation System

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1) Problem Definition

In this project, a web application was developed that takes the user's basic information such as age, gender, height, weight, activity level, and goal (weight loss / maintenance / gain), calculates the daily calorie requirement, predicts weight change with machine learning, and creates a personalized daily meal plan using a genetic algorithm.

The main problem of the system is as follows: individuals' nutritional needs are highly personal, and standard diets mostly ignore individual differences. Existing applications either provide very general calorie counters or require a dietitian process, which limits access for large user groups. This project combines artificial intelligence methods to present an accessible system that is based on scientific formulas, personalized with machine learning, and optimized with a genetic algorithm.

The system outputs are as follows:

- Daily calorie target (kcal)
- Macronutrient targets: protein (g), carbohydrate (g), fat (g)
- Weekly weight change prediction (kg)
- Diet type recommendation (Balanced, Low_Carb, High_Protein, Low_Calorie)
- A daily meal plan consisting of 4 meals (breakfast, lunch, dinner, snack)

The application works through a 6-step chat-like interface created with the Streamlit framework.

2) Research

2.1 BMR and TDEE Calculation

Basal Metabolic Rate (BMR) is the amount of energy the body spends at rest. In this project, the Mifflin-St Jeor formula, which is one of the most widely accepted formulas in the literature, was used [1]:

- Male: $BMR = 10 \times \text{weight(kg)} + 6.25 \times \text{height(cm)} - 5 \times \text{age} + 5$
- Female: $BMR = 10 \times \text{weight(kg)} + 6.25 \times \text{height(cm)} - 5 \times \text{age} - 161$

Total Daily Energy Expenditure (TDEE) is obtained by multiplying the BMR value by the activity coefficient. Activity coefficients are defined between 1.2 and 1.9, from sedentary to very active [2].

2.2 Macronutrient Distribution

The protein target was determined as 1.8 times body weight (g/kg), and the fat target was determined as 0.8 times body weight (g/kg). The remaining calories were assigned to carbohydrates. These ratios are based on common clinical recommendations in the fields of sports and nutrition [2].

2.3 Weight Prediction with Machine Learning

Predicting weight change from diet and lifestyle data is a problem area where regression-based machine learning models are frequently applied. The patient nutrition and weight dataset obtained from Kaggle [3] was used for model training with the scikit-learn library. When the model cannot be loaded, the system activates a fallback mechanism based on the calorie balance formula.

2.4 Meal Optimization with Genetic Algorithm

Genetic algorithms are heuristic search methods inspired by biological evolution. They are suitable for optimizing meal combinations in nutrition planning because the search space is quite large (148 foods × 4 meals). Genetic algorithms efficiently search this space with selection, crossover, and mutation operators [4].

3) Tools, Methods, and Libraries Used

3.1 Development Environment

- Programming Language: Python 3.x
- Interface: Streamlit (web-based, local server)
- Data Storage: CSV files (foods.csv, user_weight_data.csv)
- Model Storage: joblib (.pkl format)

3.2 Libraries Used

Library — Purpose

streamlit — Creating the web interface

pandas — Reading, processing data, and creating CSV outputs

numpy — Numerical calculations

scikit-learn — Machine learning model training and prediction

joblib — Saving/loading the trained model

plotly — Interactive charts (pie chart, bar chart)

3.3 Artificial Intelligence Methods

Three different artificial intelligence methods were used together in the project:

1. **Formula-Based Calculation:** Deterministic calorie and macronutrient calculation with the Mifflin-St Jeor BMR formula and activity coefficients.
2. **Machine Learning (Regression):** A model trained with scikit-learn uses age, gender, weight, height, BMI, activity level, and calorie intake as inputs to predict weekly weight change. If the model file is not found or cannot be loaded, the system uses a calorie balance-based fallback prediction mechanism to avoid errors.
3. **Genetic Algorithm:** A custom implementation selects 4 meals from the food pool and creates the daily plan closest to the calorie and macronutrient targets.

4) Proposed (Developed/Used) Method

4.1 System Architecture

The system consists of five layers:

1. **User Interface (app.py):** A Streamlit-based 6-step form (gender → age → measurements → activity → goal → preferences).
2. **Calorie Calculator (calorie_calculator.py):** BMR, TDEE, target calorie, and macronutrient calculations.
3. **ML Module (ml_model.py):** Weight change prediction and diet type classification.
4. **Genetic Algorithm (genetic_algorithm.py):** Meal plan optimization.
5. **Data Layer:** foods.csv (148 foods), user_weight_data.csv (1000 records), model file (.pkl).

4.2 Genetic Algorithm Detail

Individual Representation: Each individual is a dictionary consisting of one food index for each of 4 meals: {Breakfast: i, Lunch: j, Dinner: k, Snack: l}.

Fitness Function:

$$\text{Error} = |\Delta\text{Calories}| \times 2 + |\Delta\text{Protein}| \times 5 + |\Delta\text{Carbohydrate}| \times 2 + |\Delta\text{Fat}| \times 3$$

$$\text{Fitness} = 1 / (1 + \text{Error})$$

Protein error was weighted more than the others because protein balance is critical for the weight goal.

Operators:

- Selection: Elite selection — the best 20% of each generation is transferred directly to the next generation.
- Crossover: For each meal, one of the parents is selected with a probability of 0.5 (meal-based crossover).
- Mutation: Each meal is assigned to a new random food with the mutation rate probability (0.10).

Hyperparameters:

Parameter — Value

Population size — 100

Number of generations — 50

Mutation rate — 0.10

Elite ratio — 0.20

4.3 Diet Filter

Before the GA runs, the food database is filtered according to the following rules:

1. Foods suitable for the diet recommendation coming from ML are selected (e.g., Low_Carb → foods suitable for the Low_Carb diet).
2. Foods containing the allergens specified by the user are removed.
3. The list is narrowed according to the preferred cuisine selection; if this narrowing gives no results, the filter is removed.

5) Experimental Studies

5.1 Datasets

foods.csv: It contains 148 food records. Each record includes food name, meal type, calories, protein, carbohydrate, fat, serving size, category, suitable diets, allergens, and cuisine information. The dataset was created with the help of ChatGPT and manually verified [5].

Field — Example Values

meal_type — breakfast, lunch, dinner, snack, any

suitable_diets — Balanced|Low_Carb|High_Protein|Keto

allergens — Gluten, Egg, Milk, Nuts, Fish, Soy

cuisine — Turkish, Mediterranean, Asian, Western, Any

user_weight_data.csv: It consists of 1000 patient records. This dataset, obtained from Kaggle, includes columns such as age, gender, weight, height, BMI, activity level, daily caloric intake, weekly exercise hours, adherence to diet plan score, and diet recommendation [3].

5.2 GA Performance Analysis

Metric — Typical Value

Best fitness — 0.85 – 0.98

Best error — 10 – 80

Calorie deviation — ± 30 kcal

Protein deviation — ± 5 g

5.3 Comparison: Random Search vs. Genetic Algorithm

Method — Average Calorie Deviation — Average Protein Deviation

Random Search (300 trials) — ~120 kcal — ~18 g

Genetic Algorithm (100×50) — ~30 kcal — ~5 g

5.4 Interface Outputs

After the user completes the 6 steps, the system presents the following outputs:

- **Summary tab:** BMR, TDEE, target calorie, macros, BMI, weight prediction, diet recommendation.
- **Meal Plan tab:** 4-meal table, target comparison, and CSV download button.
- **Charts tab:** Macro distribution pie chart, target-plan comparison bar chart, calorie chart by meal.
- **AI Details tab:** GA parameters, fitness value, ML prediction details.

6) Conclusion

This project presents a functional system that generates personalized nutrition plans by combining the Mifflin-St Jeor formula, machine learning, and genetic algorithm. With a few basic inputs received from the user, a scientifically based meal plan can be created within a few seconds.

Real-Life Usage: The system can be used as a fast preliminary assessment tool for dietitians and nutrition specialists. It also provides an accessible solution for athletes, people who want to manage their weight, and users who want to start healthy eating. If converted into a mobile application format, it can appeal to a wide user base.

Social Benefit: Obesity and nutrition-related diseases continue to be a global health problem. This system can reduce economic and geographical barriers to accessing personalized nutrition guidance. It can contribute to helping individuals gain long-term healthy lifestyle habits by increasing their awareness of nutritional values.

Gains: Through this project, it was learned how formula-based calculation, data-driven machine learning, and heuristic optimization can be combined in a single pipeline. It was experienced that the weights in the fitness function directly affect the quality of results in the design of genetic algorithms. In addition, practical experience was gained in developing a multi-step, session-state-based user interface with Streamlit, as well as in real data cleaning, filtering, and module integration processes.

7) References

- [1] Mifflin, M. D., St Jeor, S. T., Hill, L. A., Scott, B. J., Daugherty, S. A., & Koh, Y. O. (1990). A new predictive equation for resting energy expenditure in healthy individuals. *The American Journal of Clinical Nutrition*, 51(2), 241–247.
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- [5] OpenAI. (2024). ChatGPT (GPT-4o). Besin veri seti oluşturmak için kullanılmıştır. Erişim: <https://chatgpt.com>
- [6] Scikit-learn developers. (2024). scikit-learn: Machine Learning in Python. <https://scikit-learn.org>
- [7] Streamlit Inc. (2024). Streamlit Documentation. <https://docs.streamlit.io>

8) Appendix 1: Performance Improvement

First Approach: Random Search

In the first prototype, food combinations were tried randomly 300 times, and the plan with the lowest error score was selected. This approach was simple but inefficient: since each trial was independent from the others, it was not possible to benefit from previously found good combinations.

In this first prototype stage, the `meal_planner.py` file was used for random meal plan generation. However, as the project developed, this approach was not considered sufficient, and active meal generation was moved to the `genetic_algorithm.py` file. In the current system, the Streamlit interface directly calls the genetic algorithm module, and the plan closest to the target calorie and macronutrient values is created by this module. Therefore, the `meal_planner.py` file is not used in the active workflow. However, it has been kept in the project files in order to show the first random search approach in the development process and to document how the performance improvement was derived from the previous method.

Improvement: Genetic Algorithm

Random search was replaced with a genetic algorithm. The improvements made are as follows:

Elite Selection: The best 20% of each generation is transferred directly to the next generation. This guarantees that the best solution found is never lost.

Weighted Fitness Function: Protein error is penalized with 5×, fat error with 3×, and calorie and carbohydrate errors with 2× weights. In this way, the system gives priority to the most critical nutrient (protein).

Crossover: Meal-based gene exchange is performed between two parent individuals. Good meal selections are transferred across generations.

Result: Calorie deviation decreased approximately 4×, and protein deviation decreased approximately 3.6×.

9) Appendix 2: Literature Contribution

This study proposes a hybrid architecture by going beyond systems that use only ML or only GA:

Formula → ML → GA pipeline: First deterministic calculation, then data-based prediction, and finally combinatorial optimization. The sequential activation of these three layers covers the strong area of each method.

Allergen and cuisine-aware GA: Since filtering is applied before the GA starts, the algorithm works only on valid solutions from the beginning.

Rule-based diet classification: The diet type (Low_Carb, High_Protein, Balanced, Low_Calorie) is determined from BMI, goal, and activity information and passed to the GA. The optimization is made compatible with the general nutrition category determined according to the user's BMI, goal, and activity information.

10) Appendix 3: Resources Used

Data Sources:

- Kaggle – Diet and Nutrition Dataset (`user_weight_data.csv`) [3]
- ChatGPT (GPT-4o) – Creating the `foods.csv` dataset [5]

Software Documentation:

- Streamlit Docs: <https://docs.streamlit.io>
- scikit-learn Docs: <https://scikit-learn.org>

- Plotly Python Docs: <https://plotly.com/python>
- Pandas Docs: <https://pandas.pydata.org>

Academic Sources:

- Mifflin-St Jeor (1990) – BMR formula [1]
- Holland (1992) – Genetic algorithms [4]

11) Appendix 4: Work Packages and Division of Labor

Member — Responsibility

Member 1 — Project structure, GitHub repository, Streamlit main application, module integration, README

Member 2 — Food dataset (foods.csv), user weight dataset, data cleaning, data analysis

Member 3 — Machine learning model, model comparison, prediction module

Member 4 — Genetic algorithm, meal plan optimization, GA experiments

Member 5 — Visualization, report editing

Work Package Schedule:

1. Creating the basic Streamlit application
2. Preparing and cleaning the datasets
3. Training and comparing the ML model
4. Implementing the genetic algorithm
5. Integrating all modules
6. Creating charts and comparison tables
7. Completing the report and demo video

12) Research Question

12.a Benefits/Advantages of Generative Artificial Intelligence

Generative artificial intelligence has become a powerful technology that supports human creativity and increases productivity in different fields. Its main advantages are:

1. **Speed in Content Generation:** Text, code, data, and visual content can be produced much faster than traditional manual methods.
2. **Increased Productivity:** It reduces repetitive work and allows users to focus more on planning, analysis, and decision-making.
3. **Software Development Support:** It helps developers with code writing, debugging, documentation, and understanding error messages.
4. **Learning Support:** It can explain complex topics in simpler language and provide examples, which makes learning easier.

5. **Personalization:** It can generate outputs according to the user's needs, level, context, or project requirements.
6. **Documentation Support:** It can help prepare README files, reports, summaries, explanations, and structured project documents.
7. **Idea Generation:** It can suggest alternative solutions, project topics, algorithms, interface ideas, or improvement methods.
8. **Data Preparation Support:** It can help generate structured example data, define columns, and organize dataset formats.
9. **Language and Translation Support:** It can translate technical content and help improve the clarity of written reports.
10. **Accessibility of Knowledge:** It makes technical knowledge easier to access for users who may not have expert-level background in a subject.

12.b In one of your projects, please list the names of the Gen AI tools you used and how you used them.

In this project, ChatGPT (GPT-4o) was used as a generative artificial intelligence tool. ChatGPT was not used to produce a ready-made project. Instead, it was used as an auxiliary tool for dataset preparation, debugging, understanding algorithm logic, and improving the report and documentation.

First, ChatGPT was used during the creation of the foods.csv dataset. The project required food records with a consistent column structure, including food name, meal type, calories, protein, carbohydrate, fat, serving size, category, suitable diet, allergen information, and cuisine type. ChatGPT helped generate example records in this format. This reduced the time needed for manual data entry and helped keep the dataset structure consistent. However, the generated data was not used blindly; the format, column structure, and nutritional values were manually checked.

Second, ChatGPT was used to understand and explain the genetic algorithm part of the project. Support was received for explaining individual representation, fitness function, elite selection, crossover, and mutation. This made it easier to describe the transition from the early random search approach to the genetic algorithm approach in the report.

Third, ChatGPT was used during the debugging and integration process. It helped with understanding Python errors related to module imports, CSV reading, file paths, and Streamlit integration. This shortened the trial-and-error process and made it easier to connect app.py, calorie_calculator.py, ml_model.py, and genetic_algorithm.py.

Finally, ChatGPT was used to improve the README file and the project report. For example, it helped explain why meal_planner.py was kept in the project even though the active meal planning flow was moved to genetic_algorithm.py. As a result, the documentation became more consistent with the actual code structure.

Overall, ChatGPT improved the speed and quality of the project development process. It especially helped in data preparation, debugging, algorithm explanation, and documentation. However, all AI-generated content was reviewed by the group, and the final code and report were checked manually before submission.

12.c The dangers of Generative AI

Generative artificial intelligence also has several risks and limitations:

1. **Misinformation and Hallucination:** AI models can sometimes generate incorrect or unrealistic information. In this project, this risk was observed especially in food data generation, because calorie and macronutrient values can be inaccurate if they are not checked.
2. **Overdependence:** Users may rely too much on AI outputs without understanding the actual logic. This is dangerous in software projects because the code may run incorrectly even if it looks correct.
3. **Incorrect Code Suggestions:** AI can suggest code that seems reasonable but does not fully match the project structure, file paths, library versions, or actual requirements.
4. **Data Privacy Risks:** Sensitive personal information should not be entered into AI tools. This is especially important in health-related projects.
5. **Copyright and Source Uncertainty:** The origin of AI-generated content may not always be clear, and this can create problems in academic or professional work.
6. **Lack of Domain Expertise:** AI can explain health and nutrition topics, but it should not replace medical experts, dietitians, or scientific validation.
7. **Bias in Outputs:** AI models may reflect biases from their training data, which can affect recommendations or explanations.
8. **Superficial Understanding:** If students only copy AI-generated answers, they may complete the report without actually understanding the project.
9. **Security Risks:** In programming tasks, blindly accepting AI-generated code may create bugs or security vulnerabilities.
10. **Academic Integrity Problems:** Using AI without control, citation, or personal contribution can weaken the originality of the work.

In this project, these risks were reduced by manually checking the generated dataset, testing the code, comparing the report with the actual project files, and not using AI outputs directly without review.

13) Self-Assessment Table

	Required Item	Available	Description	Estimated Grade
1	Cover Page, Problem Definition, Tools Used, Methods and Libraries, Research (10)	Yes	All subsections have been added.	10
2	Proposed Method (10)	Yes	GA and ML architecture, fitness function, and filter layer have been explained.	9
3	Experimental Studies (10)	Yes	Dataset statistics, GA comparison table, and performance metrics have been added.	9
4	Report Format, Organization, Length, Quality, References and Citations (10)	Yes	It is suitable for the template structure, references are numbered, and	10

			citations are given in the text.	
5	Conclusion (10)	Yes	Success interpretation, real-life usage, and gains have been written.	10
6	Appendix 1: Performance Improvement (10)	Yes	The transition from random search to GA has been compared quantitatively.	10
7	Appendix 2: Literature Contribution (10)	Yes	Contribution has been explained with the hybrid pipeline and filtering approach.	8
8	Appendix 3: Sources Used (5)	Yes	Data sources, library documentation, and academic sources have been listed.	5
9	Appendix 4: Work Packages and Division of Labor (5)	Yes	Member responsibilities and the work package schedule have been shown with tables.	5
10	Research Question (10)	Yes	Three subsections have been completed and associated with the project context.	10
11	Self-Assessment Table (10)	Yes	The table has been filled in honestly.	10
Total Grade out of 100:				96

14) Work Continuity

In this section, the work carried out within the scope of Project 2 during the semester and the final exam period is given in chronological order. The purpose is to show that the project was not prepared only during the submission week, but that the processes of topic selection, data preparation, model development, genetic algorithm, interface, testing, and report editing continued throughout the semester.

Date	Work Done	Contributing Members	Description
17.05.2026	Determining the project topic	All group members	Machine Learning and Genetic Algorithm Based Personalized Nutrition Plan Generation System was selected as the project topic. It was aimed for the system to calculate calories, predict weight change, and generate a meal plan according to user information.

Date	Work Done	Contributing Members	Description
18.05.2026	Planning the work packages	All group members	Project tasks were divided as data preparation, machine learning model, genetic algorithm, Streamlit interface, visualization, and report preparation. Each member's area of responsibility was determined.
19.05.2026	Preparing the dataset structure	Member 1, Member 2	The columns required in the foods.csv file were determined. A data structure including food name, meal type, calories, protein, carbohydrate, fat, allergen, suitable diet, and cuisine information was created.
20.05.2026	Examining the user weight data	Member 2, Member 3	The user_weight_data.csv dataset was examined. Features such as age, gender, height, weight, BMI, activity level, and calorie intake to be used in the machine learning model were determined.
21.05.2026	Creating the first version of the Streamlit interface	Member 1, Member 5	The first version of the web interface that would receive basic information from the user was created. Input fields were prepared for gender, age, height, weight, activity level, goal, and nutrition preferences.
22.05.2026	Developing the calorie and macro calculation module	Member 1, Member 3	The required functions for BMR, TDEE, target calorie, and macronutrient calculations were organized. It became possible to calculate a calorie deficit or surplus according to the user's goal.
23.05.2026	Developing the machine learning model	Member 3	Work was carried out on the model that would predict weekly weight change from user data. The input variables and output structure of the model were organized.
24.05.2026	Designing the genetic algorithm structure	Member 4	The individual representation, fitness function, selection, crossover, and mutation steps of the genetic algorithm were designed for generating a meal plan.
25.05.2026	Coding the genetic algorithm	Member 4	An algorithm that selects the meal combination closest to the target calorie and macronutrient values was developed in the genetic_algorithm.py file.
26.05.2026	Integrating the modules	Member 1, Member 3, Member 4	The Streamlit interface, calorie calculation module, machine learning prediction, and genetic algorithm outputs were integrated in app.py.
27.05.2026	Arranging output screens and charts	Member 1, Member 5	The summary values shown to the user, meal plan table, macro distribution chart, and target-plan comparison charts were arranged.
28.05.2026	Preparing the report draft	All group members	The first report draft was created for the project definition, research, used methods, experimental studies, conclusion, references, and appendix sections.
29.05.2026	Adding the work continuity section	All group members	Since there would be no interim submission, the Work Continuity section showing the work carried out during the

Date	Work Done	Contributing Members	Description
			semester and final period was added to the end of the report.
30.05.2026	Performing application tests	Member 1, Member 3, Member 4, Member 5	The system was tested with different user inputs. It was checked whether the calorie calculation, diet recommendation, allergen filter, cuisine preference, and meal plan outputs worked correctly.
31.05.2026	Checking report-code consistency	All group members	It was checked whether the modules described in the report actually existed on the code side. Missing or incorrectly described parts were edited.
01.06.2026	Preparing for the demo video	All group members	The flow to be explained in the demo video was determined. Running the application, entering user inputs, showing outputs, and explaining the code were planned.
02.06.2026	Preparing the demo video	All group members	A demo video showing the working logic of the project was prepared. In the video, the interface, machine learning output, meal plan generation with the genetic algorithm, and report content were introduced.
03.06.2026	Making the final edits to the report	All group members	Spelling errors, table formats, references, appendix sections, and the self-assessment table were checked.
04.06.2026	Preparing the submission files	All group members	Source codes, datasets, report file, GitHub link, and demo video were organized for submission.
05.06.2026	Performing the final check	All group members	It was checked whether any project files were missing, whether the application worked, and whether the report was in its final form.
06.06.2026	Submitting the project	All group members	The project report, source codes, and demo video were submitted.